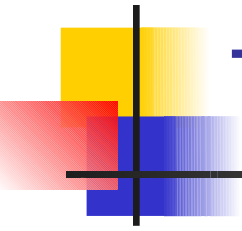


Traitement d'images

Texture

Alain Boucher - IFI



Texture

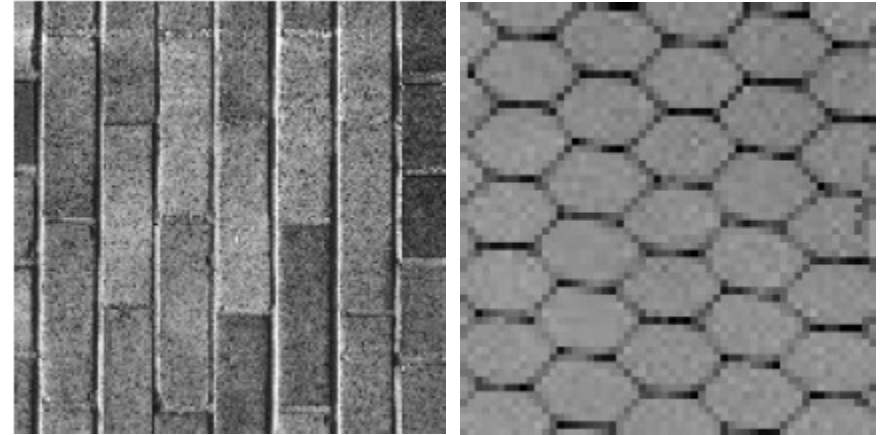
- Les textures sont
 - une information intéressante à analyser
 - mais plus difficile à extraire
- Jusqu'à présent, on définissait
 - Une **région** comme une zone homogène
 - Un **contour** comme une variation d'intensité
- Une **texture** peut être définie
 - comme une région avec des variations d'intensité
 - comme une organisation spatiale des pixels
 - ...

Exemples de textures

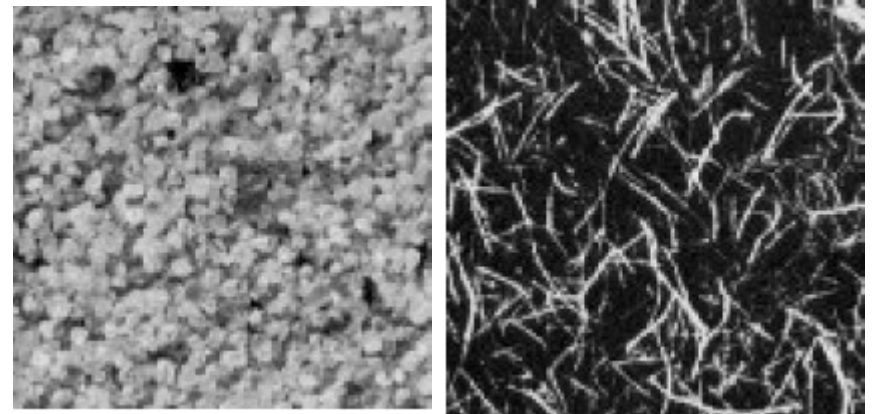


Motifs des textures

Une texture peut être **périodique** (répétition d'un motif de base) ...



...ou **non-périodique** (pas de motif de base, plus désordonnée)





Plusieurs sous-problèmes

- **Segmentation de texture**

- Diviser une image en plusieurs régions où chaque région représente une même texture

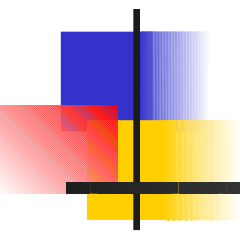
- **Synthèse de texture**

- A l'aide de petites images de texture, construire de plus grandes images en extrapolant ces textures



Analyse de la texture

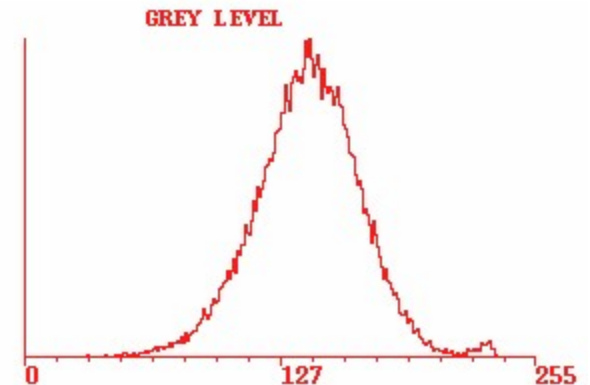
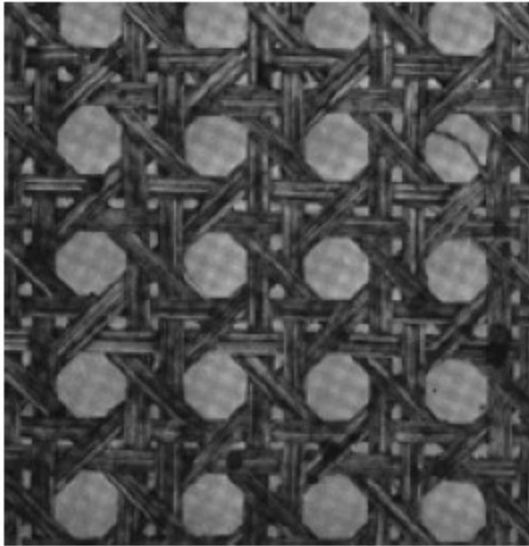
- Il existe plusieurs méthodes pour analyser la texture :
 - Statistiques du premier ordre
 - Statistiques sur l'histogramme
 - Matrices de co-occurrences
 - Recherche de motifs
 - Analyse fréquentielle
 - Filtres de Gabor
- Le plus difficile est de trouver une bonne représentation (paramètres) pour chaque texture



Analyse de la texture

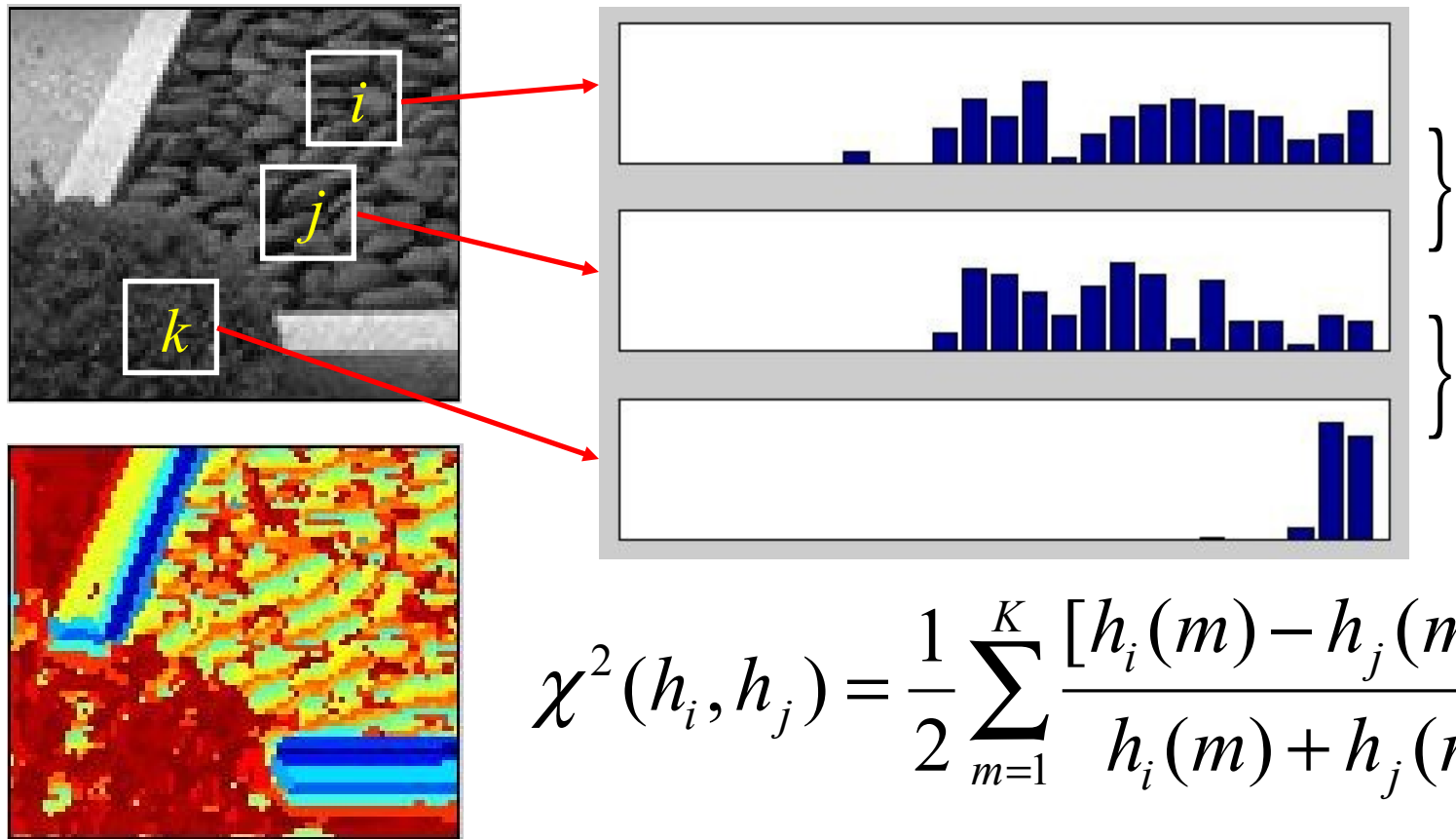
Statistiques du premier ordre

Histogrammes et textures



Distance d'histogrammes

Distance du χ^2 entre histogrammes de textures





Paramètres des textures

- On peut calculer plusieurs paramètres d'une texture à partir de l'image ou de son histogramme :

$$MOYENNE = \sum_y \sum_x g(x, y)$$

$$VARIANCE = \frac{1}{N} \sum_y \sum_x (g(x, y) - MOYENNE)^2$$

$$SKEWNESS = \frac{1}{N} \sum_y \sum_x (g(x, y) - MOYENNE)^3$$

$$KURTOSIS = \frac{1}{N} \sum_y \sum_x (g(x, y) - MOYENNE)^4$$

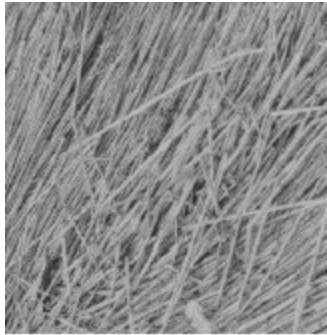
où

$g(x,y)$: niveaux de gris des pixels

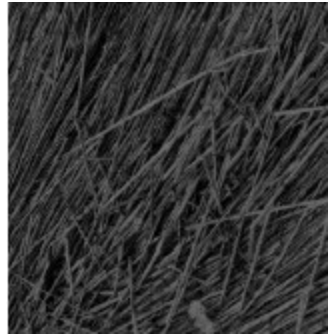
Ce sont les moments d'ordre 1,2,3,4,... de l'image.

D'autres paramètres existent (uniformité, entropie, ...)

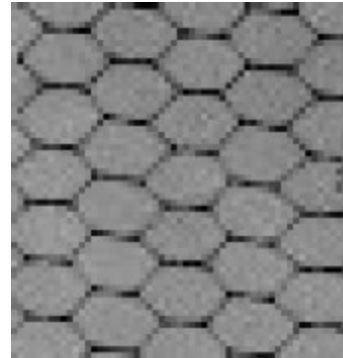
Paramètres des textures



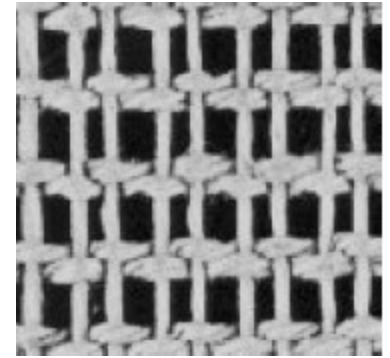
moeyenne=137



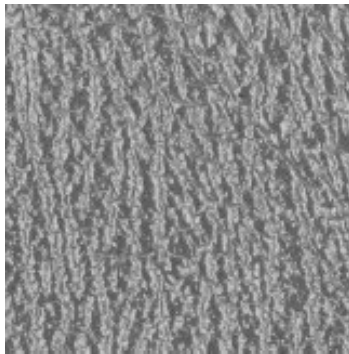
moeyenne=50



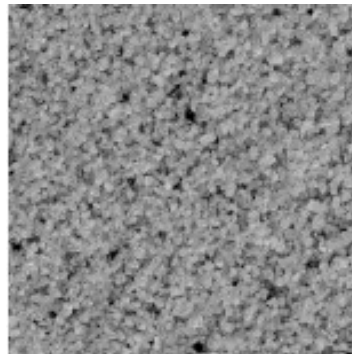
variance=30



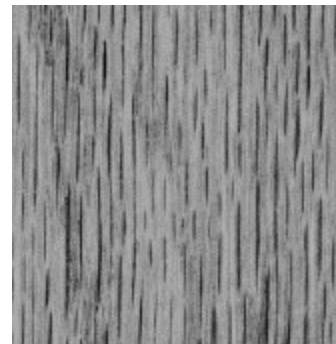
variance=71



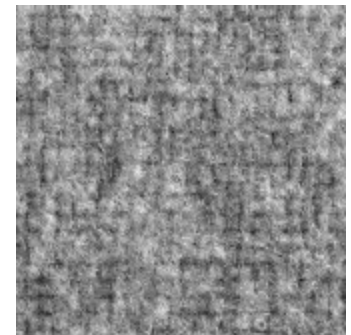
skew=0.25



skew=-1.15

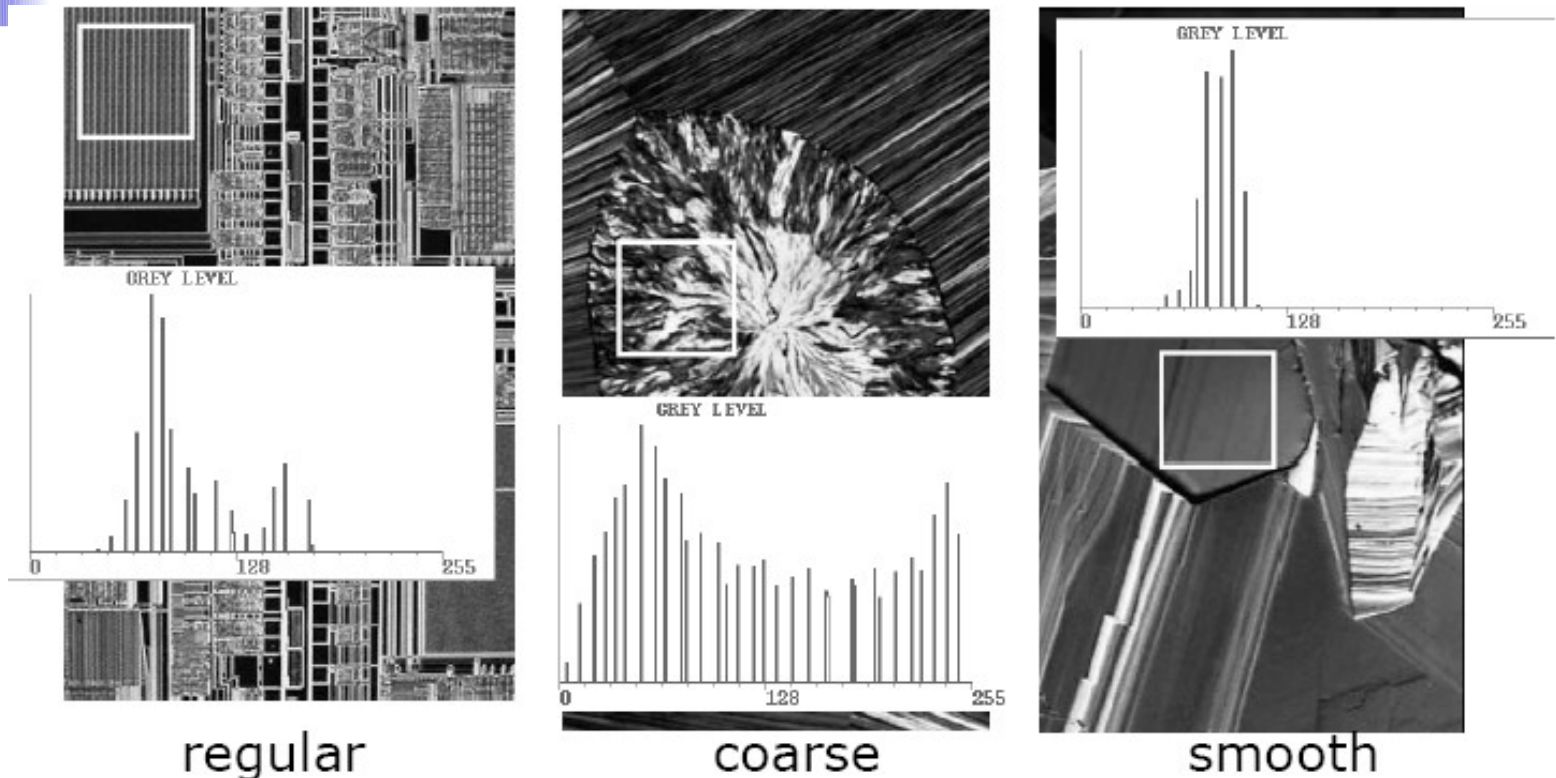


kurt=2.31

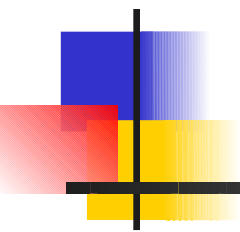


kurt=-0.09

Caractérisation des textures

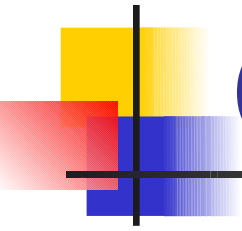


texture	m_1	σ	R_{norm}	μ_3	U	e
regular	99.72	33.73	0.017	0.750	0.013	6.674
coarse	143.56	74.63	0.079	-0.151	0.005	7.783
smooth	82.64	11.79	0.002	-0.105	0.026	5.434



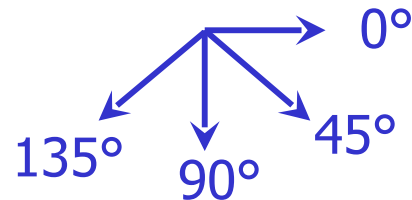
Analyse de la texture

Matrices de co-occurences



(2) Matrices de co-occurrences

- L'idée de cette méthode est d'identifier les **répétitions de niveaux de gris** selon une **distance** et une **direction**
 - *Matrices de co-occurrence (Haralick)*
- **Matrice de taille $N_g \times N_g$**
 - N_g étant le nombre de niveaux de gris de l'image (256x256)
 - On réduit souvent à des tailles 8x8, 16x16 ou 32x32
- Plusieurs matrices, pour chaque distance et direction
 - **Distance** : 1, 2, 3 (,4, ...)
 - **Direction** : 0°, 45°, 90°, 135° (, ...)
- Temps de calcul des matrices est assez long



Matrices de co-occurrences

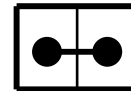
- Exemple de calcul des matrices :

1	4	4	3
4	2	3	2
1	2	1	4
1	2	2	3

Image

	1	2	3	4
1	?	?	?	?
2	?	?	?	?
3	?	?	?	?
4	?	?	?	?

*Matrice pour distance=1
et direction=0°*



On parcourt l'image et pour chaque couple de pixels formé avec la distance et la direction données, on incrémente la matrice des co-occurrences de 1

Matrices de co-occurrences

- Exemple de calcul des matrices :

1	4	4	3
4	2	3	2
1	2	1	4
1	2	2	3

Image

	1	2	3	4
1	0	0	0	1
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0

Matrice pour distance=1 et direction=0°

Couple de pixels voisins (1,4)

Matrices de co-occurrences

- Exemple de calcul des matrices :

1	4	4	3
4	2	3	2
1	2	1	4
1	2	2	3

Image

	1	2	3	4
1	0	0	0	1
2	0	0	0	0
3	0	0	0	0
4	0	0	0	1

Matrice pour distance=1 et direction=0°

Couple de pixels voisins (4,4)

Matrices de co-occurrences

- Exemple de calcul des matrices :

1	4	4	3
4	2	3	2
1	2	1	4
1	2	2	3

Image

	1	2	3	4
1	0	0	0	1
2	0	0	1	0
3	0	1	0	0
4	0	1	1	1

Matrice pour distance=1 et direction=0°

etc. (après 2 lignes)

Matrices de co-occurrences

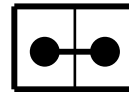
- Exemple de calcul des matrices (final) :

1	4	4	3
4	2	3	2
1	2	1	4
1	2	2	3

Image

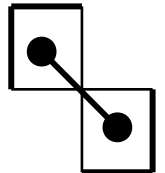
	1	2	3	4
1	0	2	0	2
2	1	1	2	0
3	0	1	0	0
4	0	1	1	1

*Matrice pour distance=1
et direction=0°*



	1	2	3	4
1	0	2	1	0
2	1	1	0	0
3	0	0	0	1
4	0	2	1	0

*Matrice pour distance=1
et direction=45°*



...et ainsi de suite pour chaque matrice (plusieurs au total)



Calculs des paramètres des matrices

- Pour chaque matrice, on peut calculer **quatorze paramètres** caractérisant la texture, dont les plus importants :

$$MOYENNE = \sum_j \sum_i p(i, j)$$

$$VARIANCE_i = \sum_j \sum_j (i - MOYENNE)^2 p(i, j)$$

$$ENERGIE = \sum_j \sum_i (p(i, j))^2$$

$$INERTIE = \sum_j \sum_i (i - j)^2 p(i, j)$$

$$ENTROPIE = - \sum_j \sum_i p(i, j) \log(p(i, j))$$

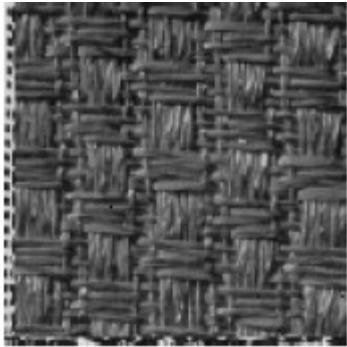
$$MOMENT \text{ DIFFERENTIEL INVERSE} = \sum_i \sum_j \frac{1}{1 + (i - j)^2} p(i, j)$$

où

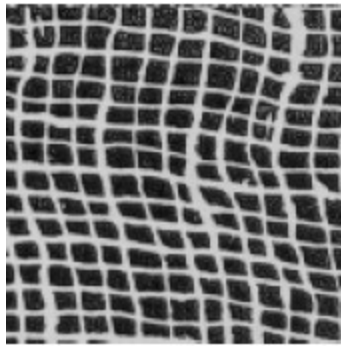
(i,j) : coordonnées dans la matrice

p(i,j) : valeurs normalisées de la matrice

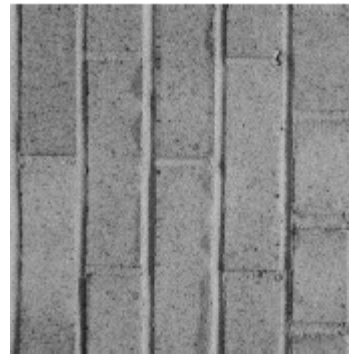
Paramètres de la matrice



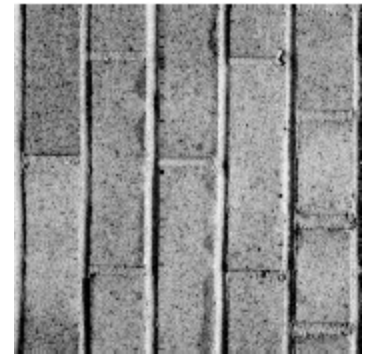
contraste=120



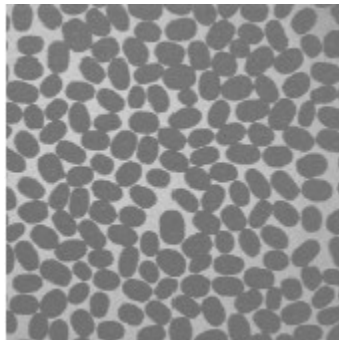
contraste=626



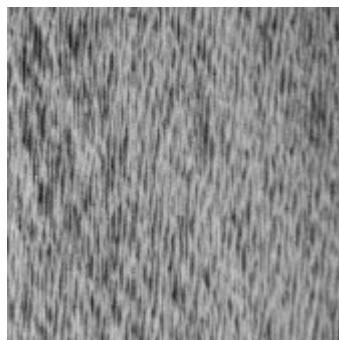
contraste=42



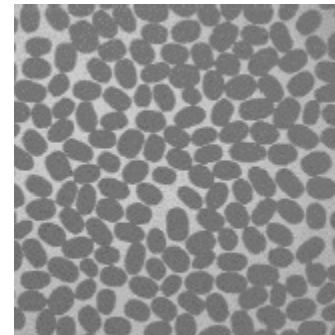
contraste=142



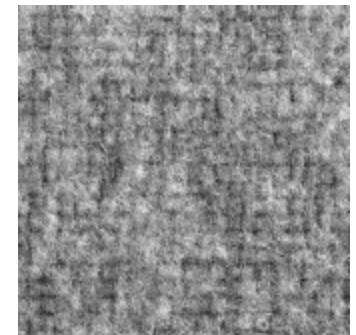
entropie=6.7



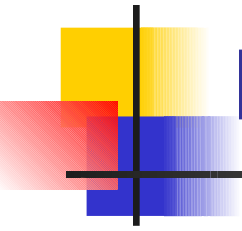
entropie=4.7



mdi=0.03



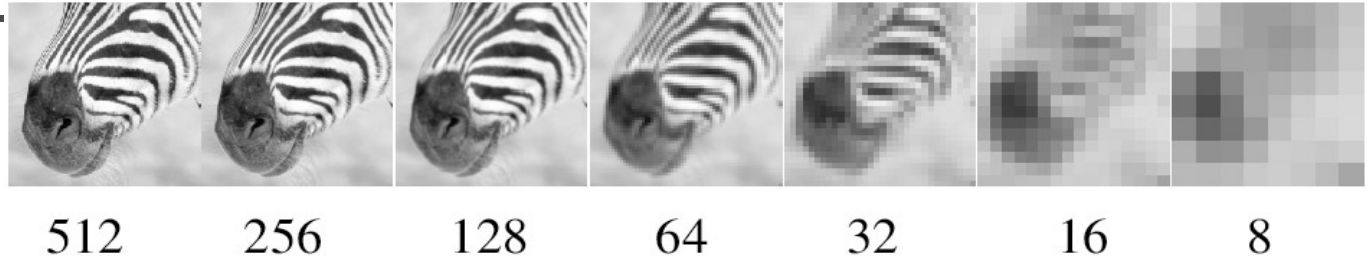
mdi=0.002



Détection des textures

- Avec les matrices de co-occurences, on espère identifier une texture particulière (selon une distance et une direction à trouver)
- Problème du calcul sur l'image entière ou sur des zones de l'image (découpage en carrés de l'image)
- Il existe plusieurs variantes de cette méthode basée sur le calcul de différences de niveaux de gris

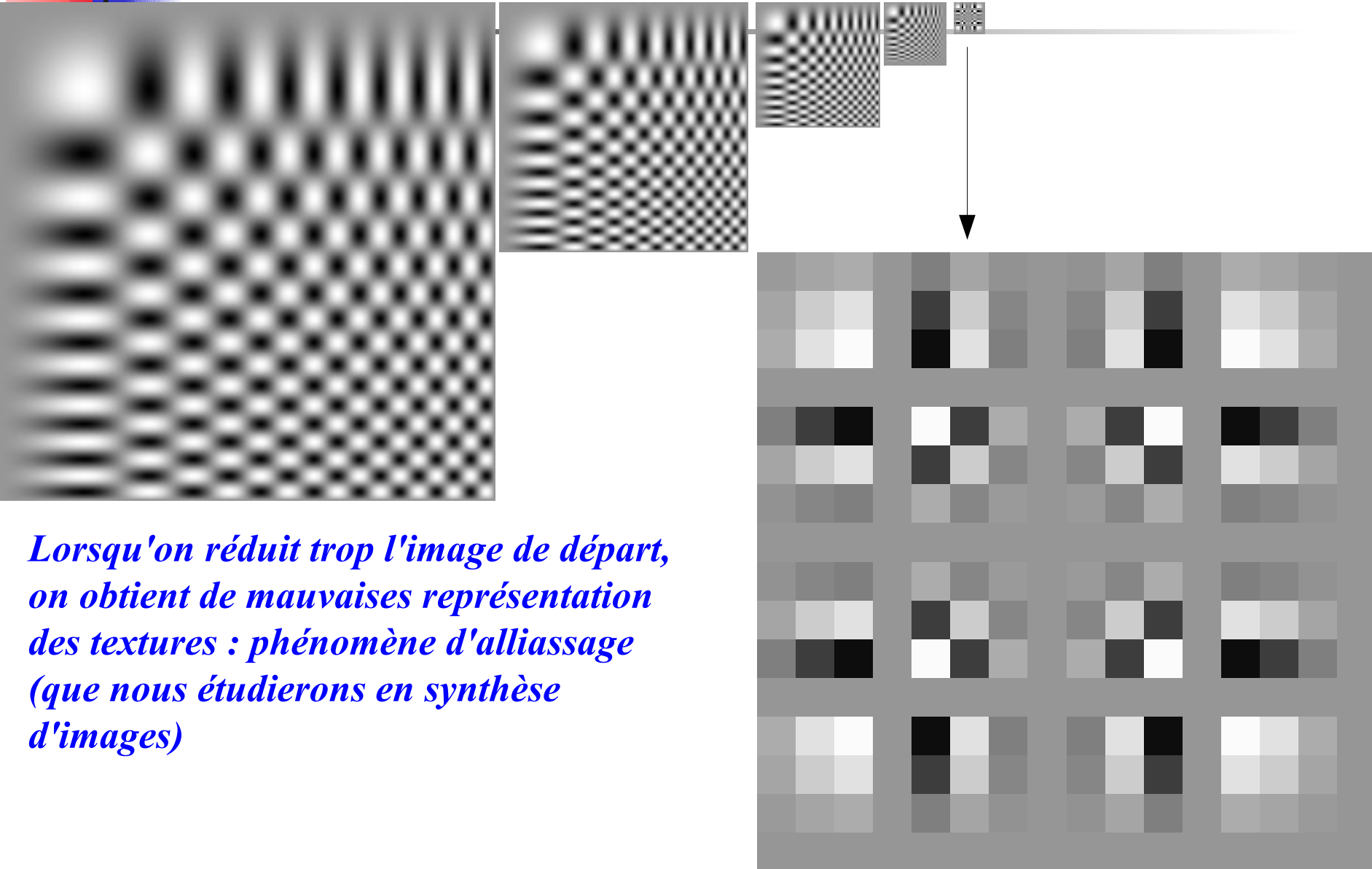
Problème d'échelle pour les textures

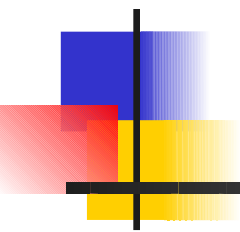


Pour bien identifier les textures, on a souvent besoin de travailler à différentes échelles...



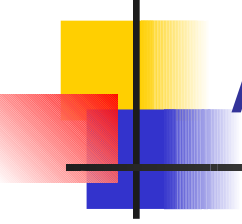
Alliassage des textures





Analyse de la texture

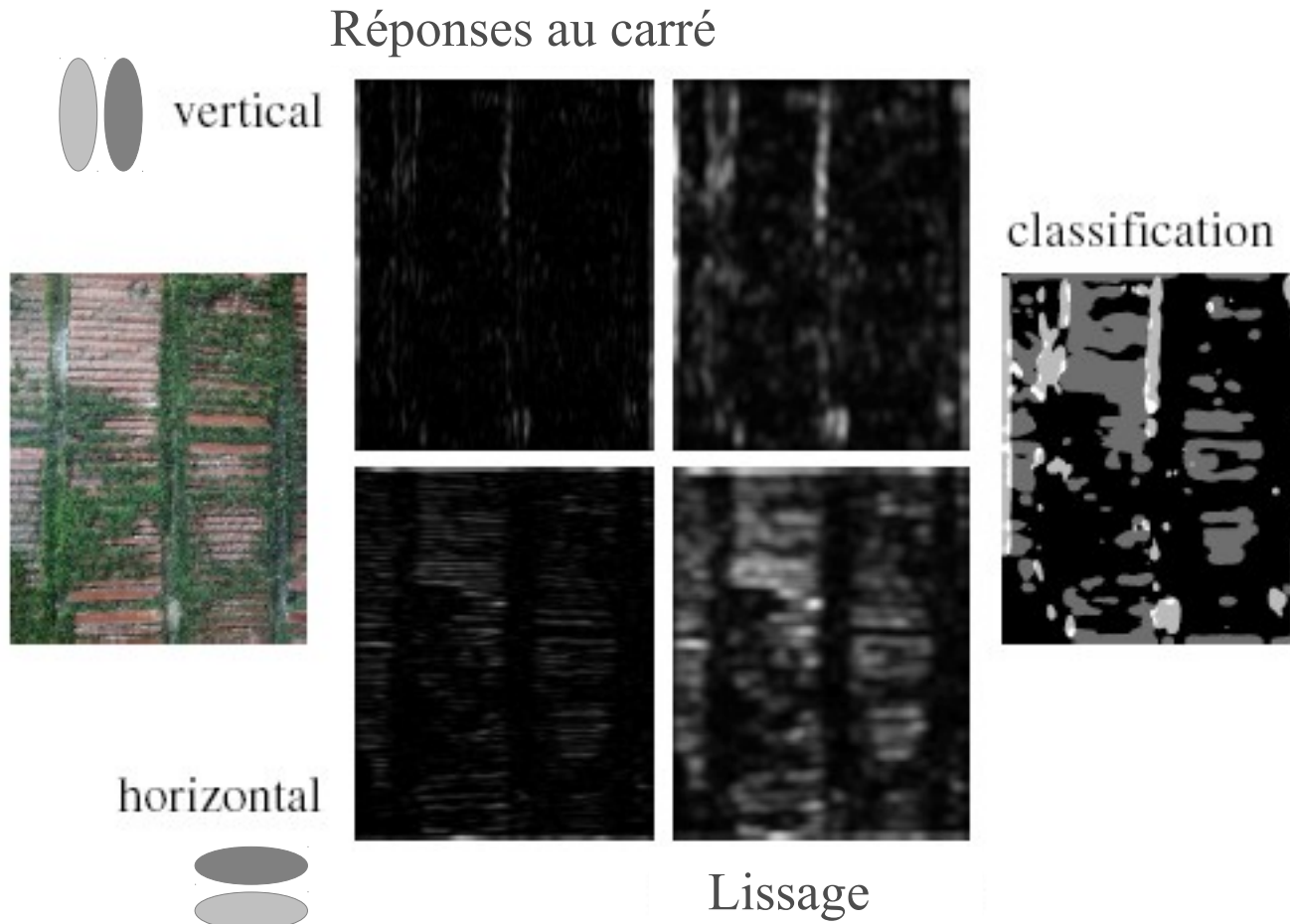
Analyse fréquentielle



Analyse fréquentielle

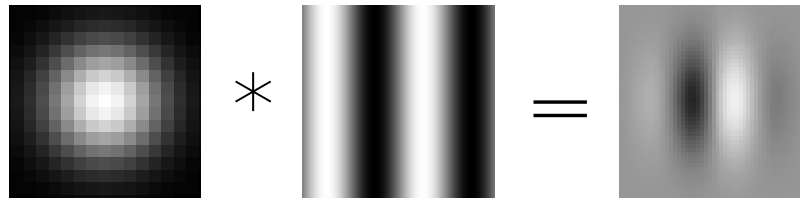
- Une texture est une répétition d'éléments avec une certaine **fréquence**
- Filtrage de l'image en utilisant des méthodes spectrales
 - Fourier
 - **Filtres de Gabor**
 - ...
- Pour rappel sur la Transformée de Fourier (FFT)
 - *(re)voir le cours de Traitement d'images*

Idée générale du filtrage fréquentiel



Filtres de Gabor

- Les filtres de Gabor sont le produit de Gaussiennes par des sinus ou des cosinus
 - Sinus / cosinus : analyse en fréquences
 - Gaussienne : différents niveaux de lissage (échelles)



Gaussienne * Sinus/Cosinus = Gabor

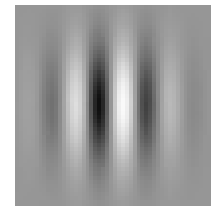
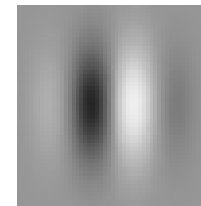
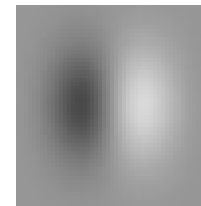
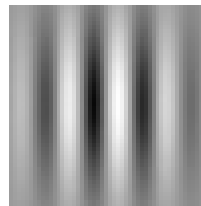
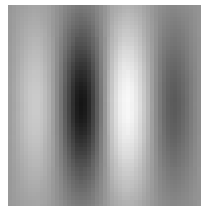
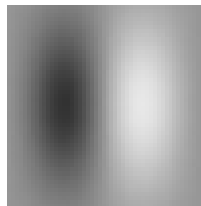
$$G_s(x, y) = \cos(\omega_x x + \omega_y y) e^{-\frac{x^2 + y^2}{2\sigma^2}}$$

$$G_a(x, y) = \sin(\omega_x x + \omega_y y) e^{-\frac{x^2 + y^2}{2\sigma^2}}$$

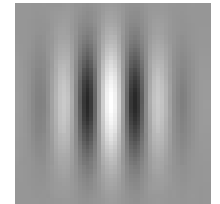
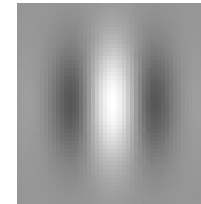
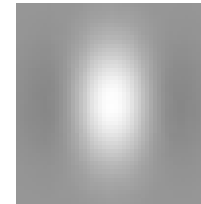
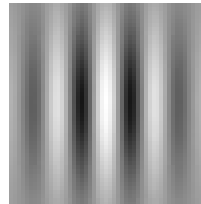
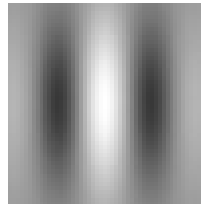
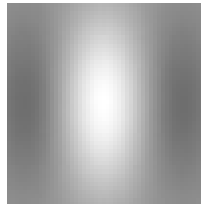
Filtres de Gabor

- Les filtres de Gabor analysent la structure
 - *à différentes échelles (fréquences)*
 - *à différentes orientations (angles)*

*Filtres
anti-symétriques
(impairs - sinus)*



*Filtres
symétriques
(pairs - cosinus)*



Grande échelle

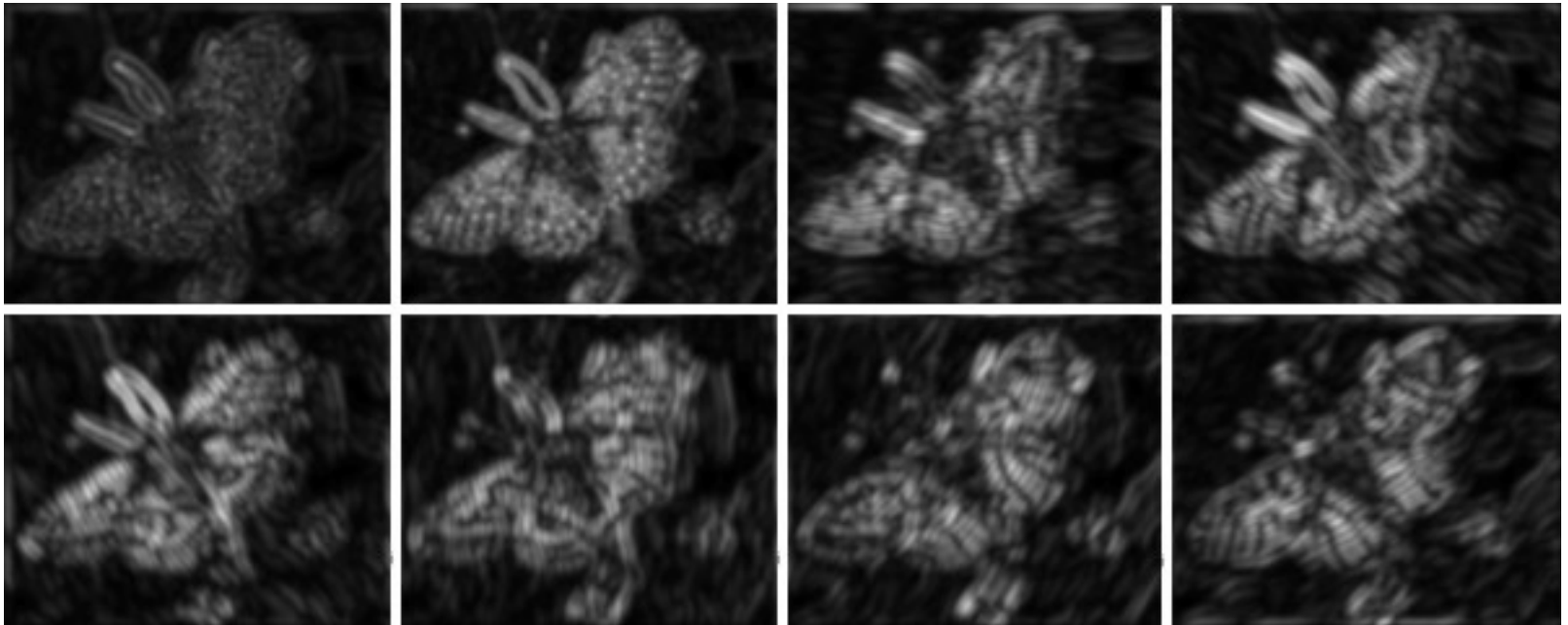
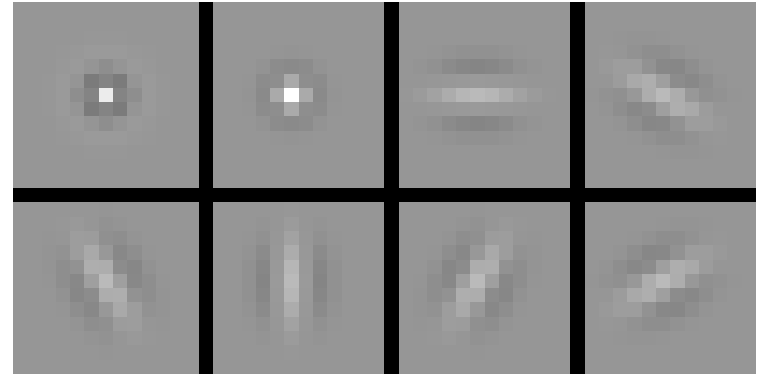
Petite échelle

Banque de filtres sur une image

*Image
d'entrée*

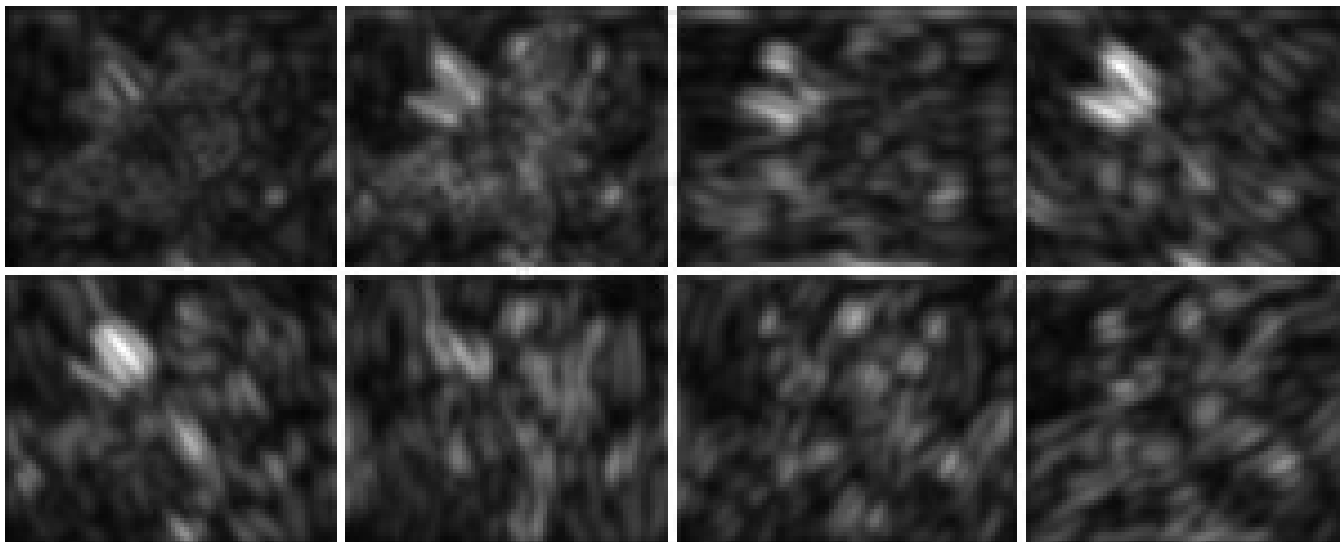
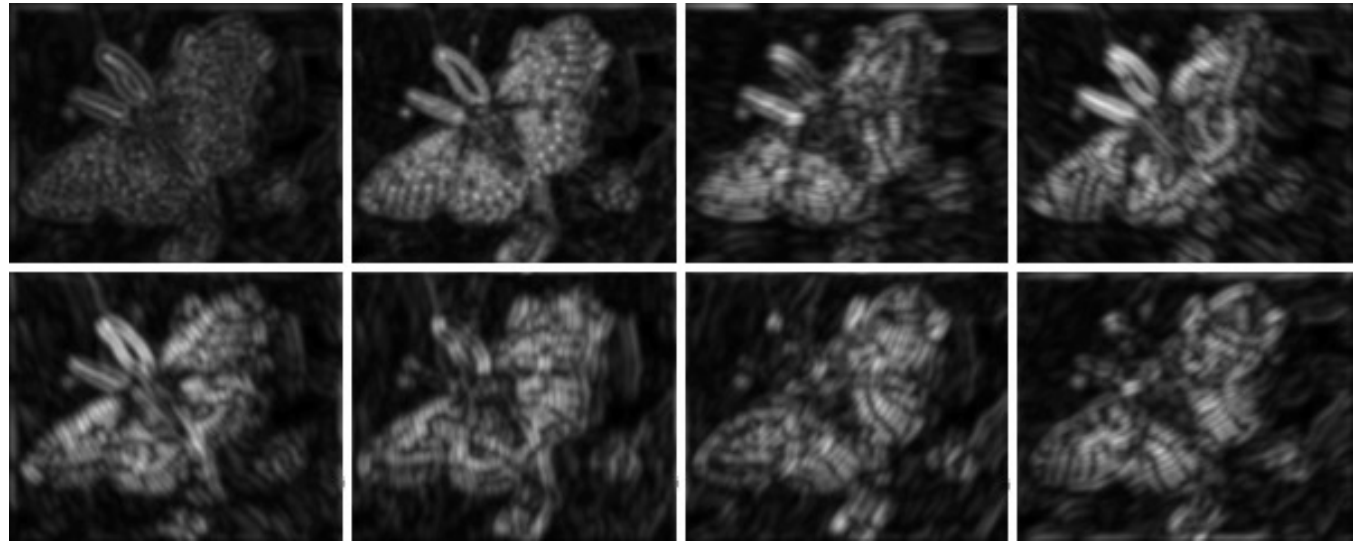


*Banque
de filtres*



Réponse des filtres à une échelle

Filtrage à différentes échelles



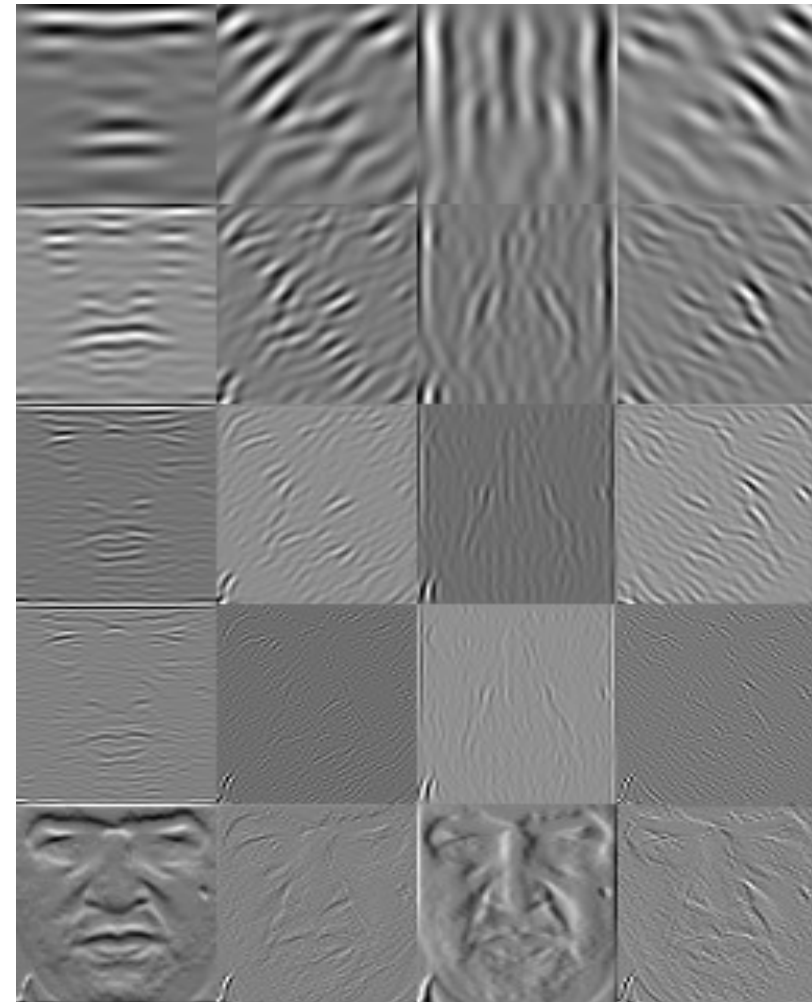
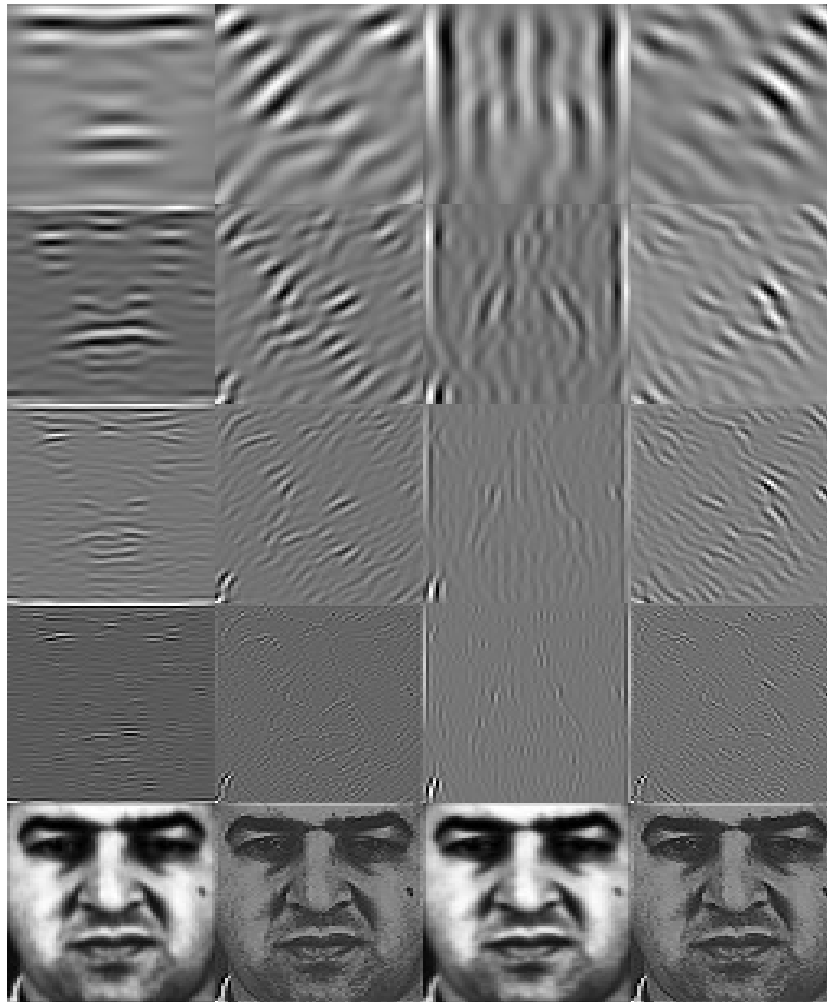
Exemple (Torch3Vision gabor2d)

Partie réelle

Partie imaginaire

(s)

*Echelle
des
filtres*



Orientation des filtres (o)



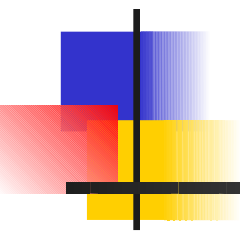
Calcul de statistiques

- On obtient une série de réponses aux différents filtres
 - *réponse complexe (partie réelle + partie imaginaire)*
 - *une série d'images de réponse pour chaque image de texture*
- Calcul de statistique sur la norme au carré
 - **$\text{norme}^2 = \text{réel}^2 + \text{imaginaire}^2$**
- Quelles statistiques faut-il calculer ?
 - Plus on a de statistiques, mieux c'est
 - Minimum : **moyenne de la norme²** et **écart-type de la norme²**



Comparaison de vecteurs

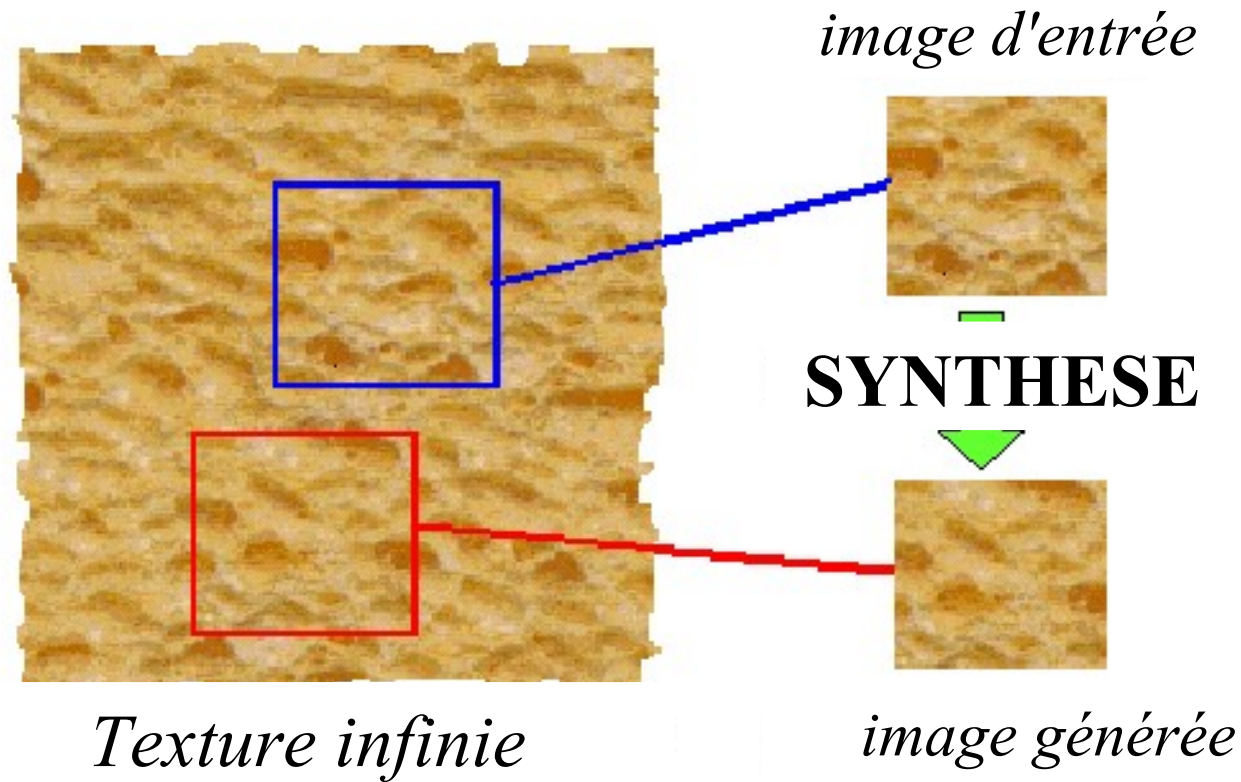
- Pour chaque image de texture, on obtient un vecteur de caractéristiques
- Comparaison de textures : calcul de distance entre les vecteurs de caractéristiques
- Distance euclidienne
 - N'est pas invariante (rotation + échelle)
- Il existe d'autres méthodes de comparaison plus efficace
 - pour obtenir l'invariance (rotation + échelle)



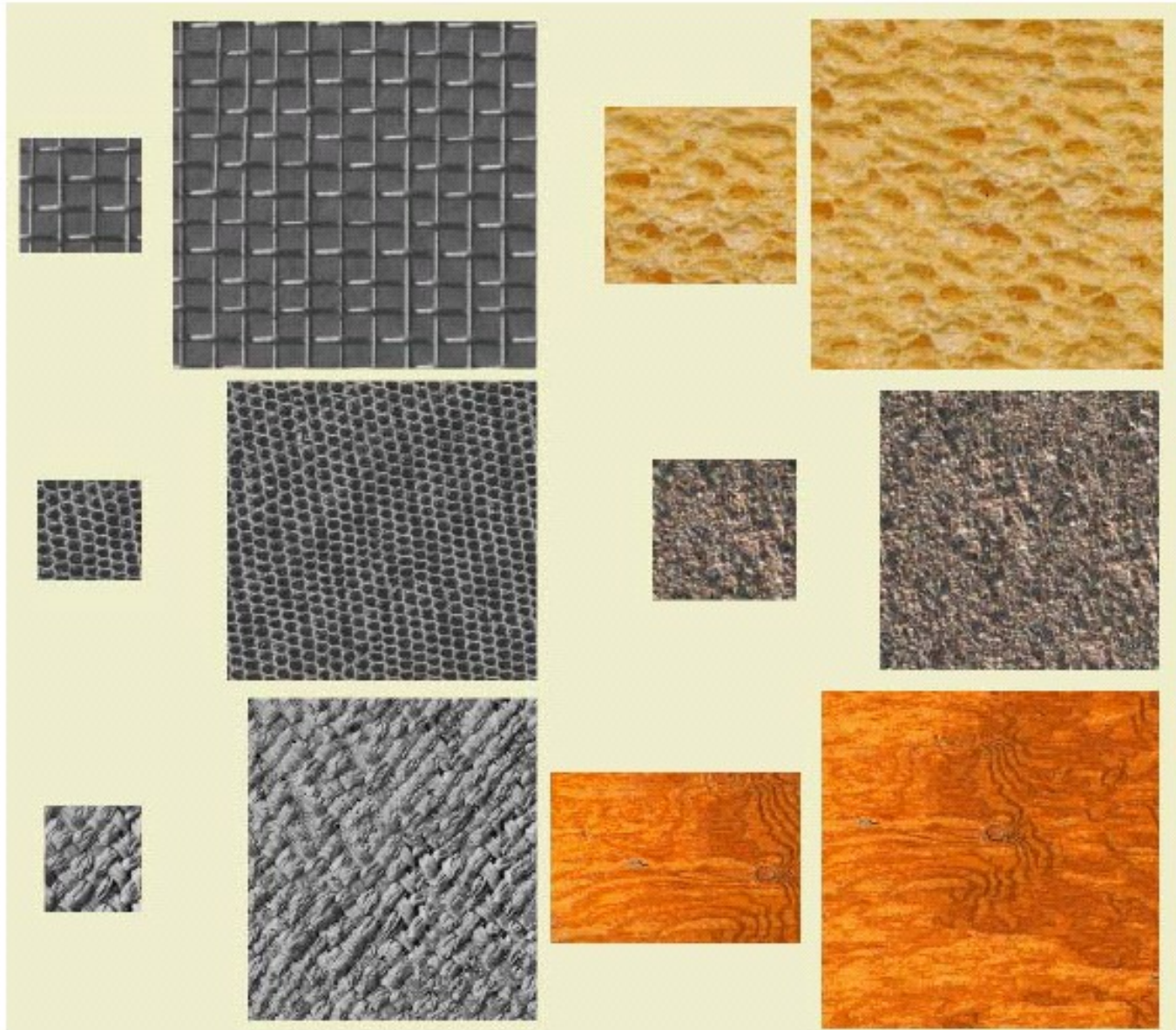
Synthèse de textures

(optionnel)

Synthèse de texture



Exemples (Efros et Leung)

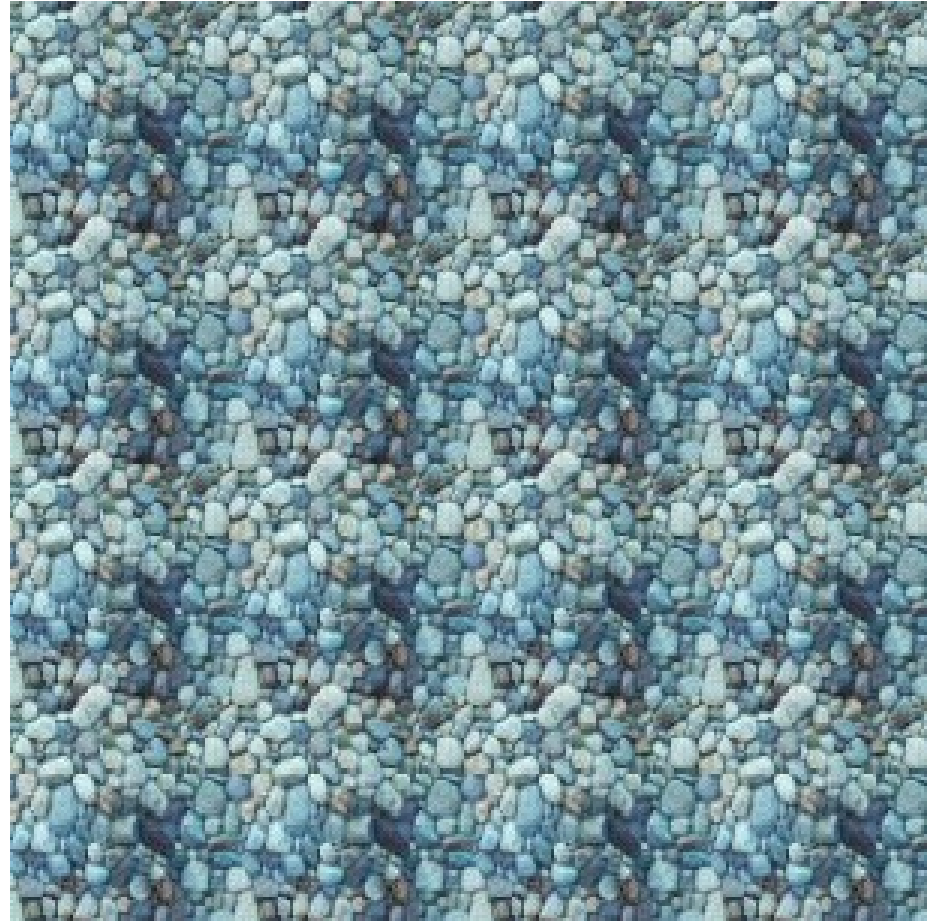


Source : David Lowe, *Texture*, CPSC 425: Computer Vision, UBC (Canada).

Pas seulement un copier-coller



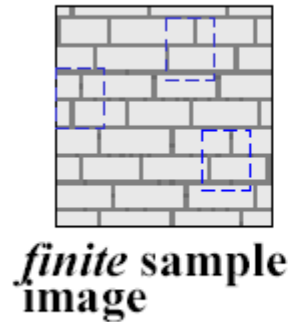
Photo



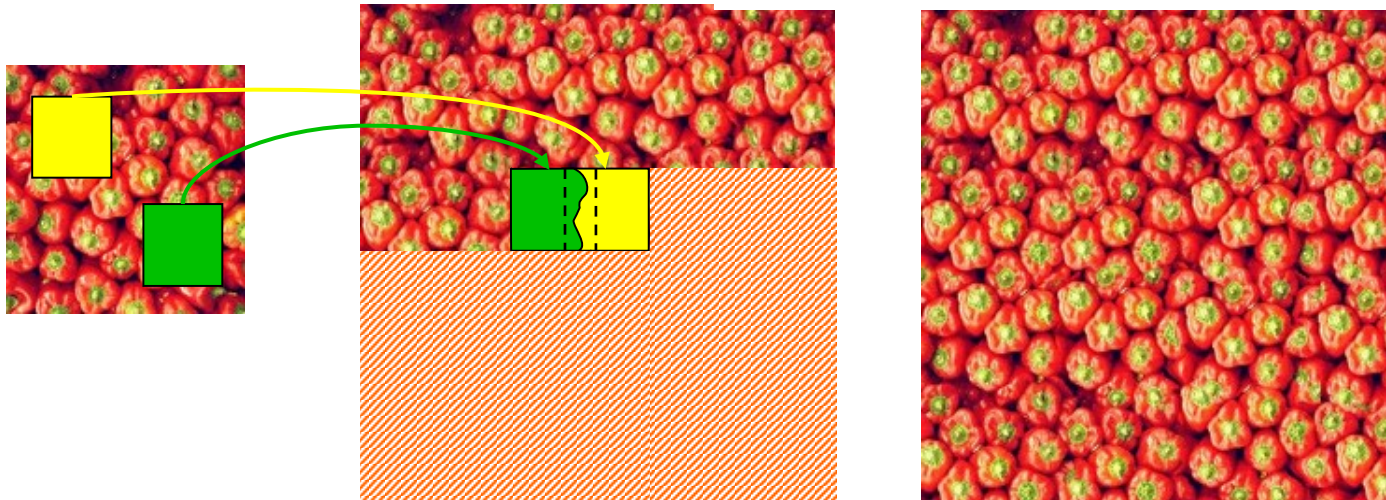
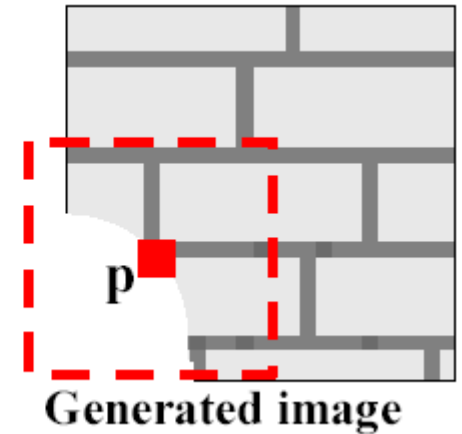
Répétition seulement

Exemples d'algorithmes

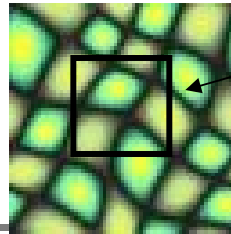
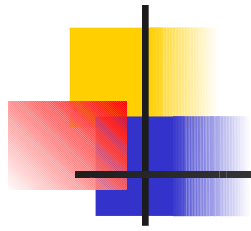
[Efros & Leung]



SAMPLE
→

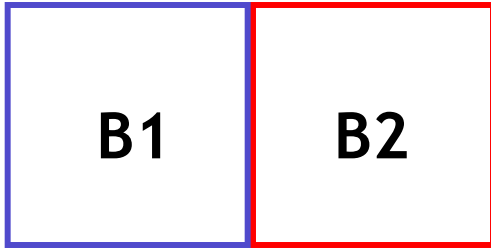


[Efros & Freeman, 2001]

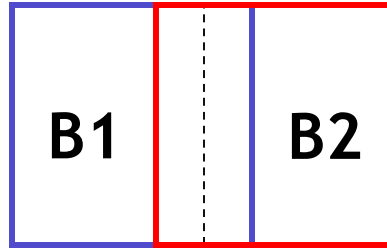


bloc

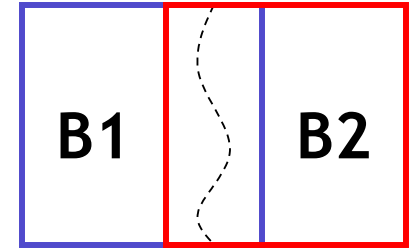
Texture d'entrée



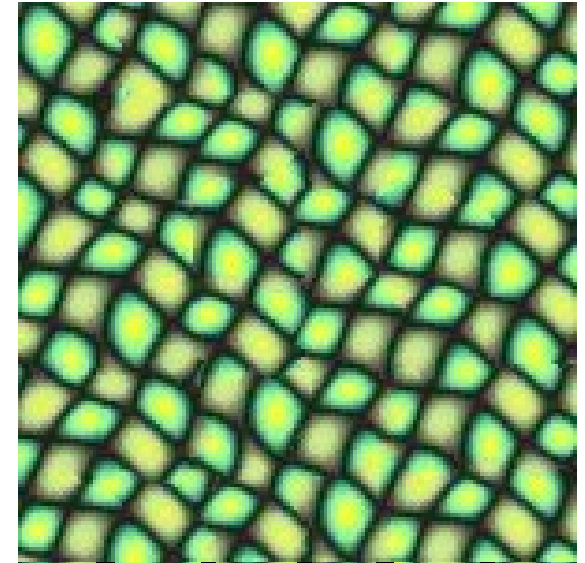
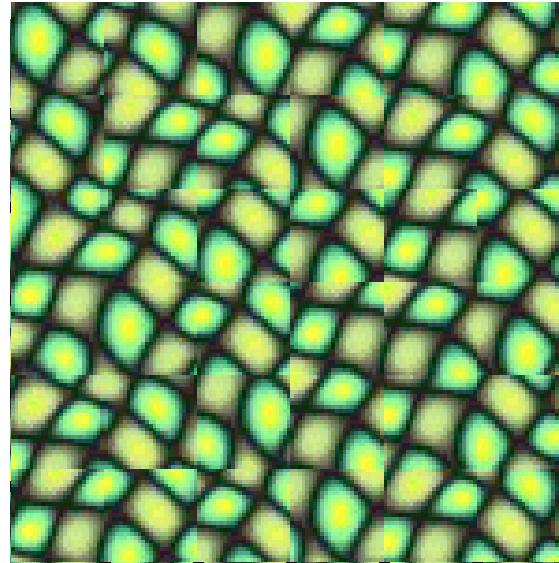
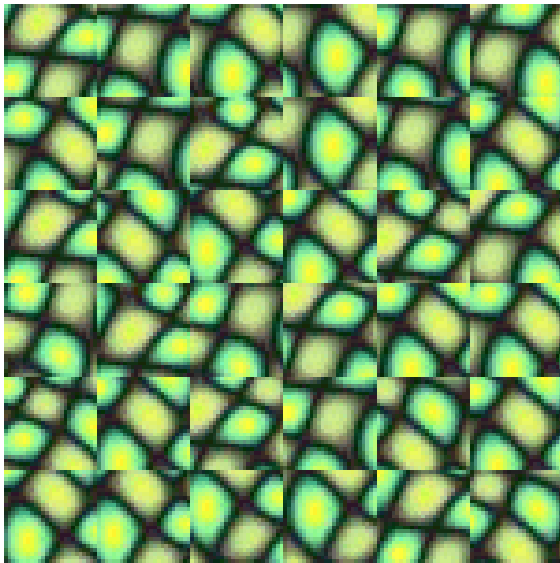
Placement aléatoire
des blocs



Recouvrement
des blocs voisins

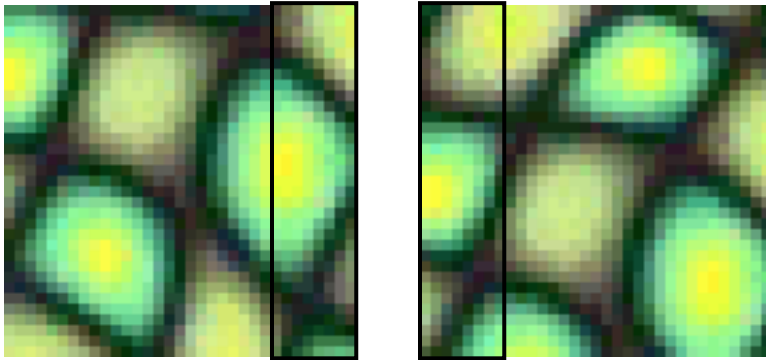


Calcul de la frontière
d'erreur minimale

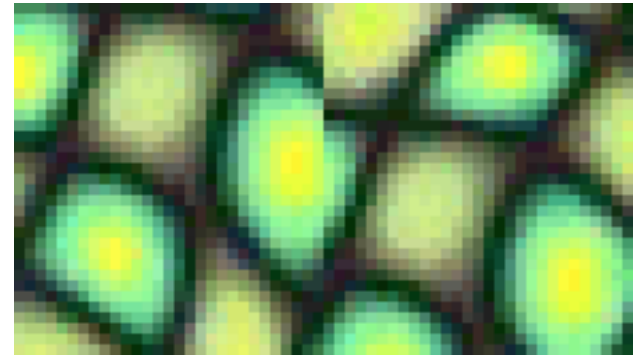


Frontière d'erreur minimale

Recouvrement de blocs

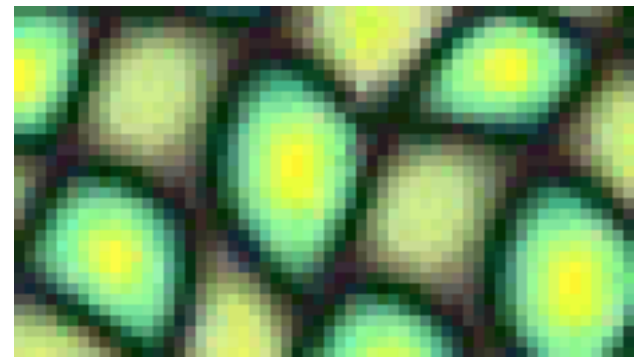


Frontière verticale



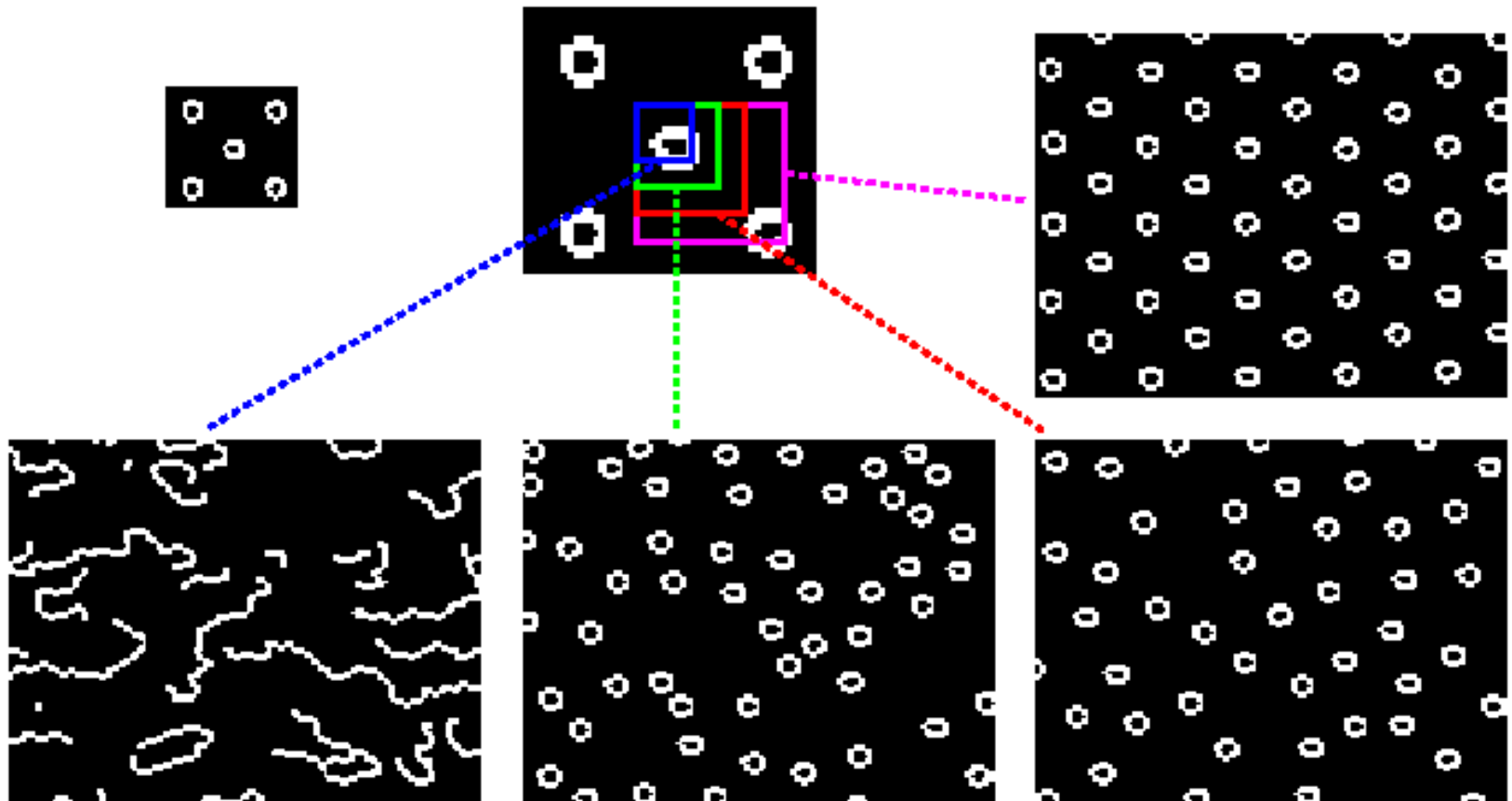
A diagram showing the calculation of the error of overlap. It consists of two vertical strips of the textured image, one on the left and one on the right, with a minus sign between them. This is followed by a large '2' as a superscript, an equals sign, and a vertical strip of the textured image with a red line indicating the boundary. Arrows point from the two strips to the '2'.

Erreur de recouvrement



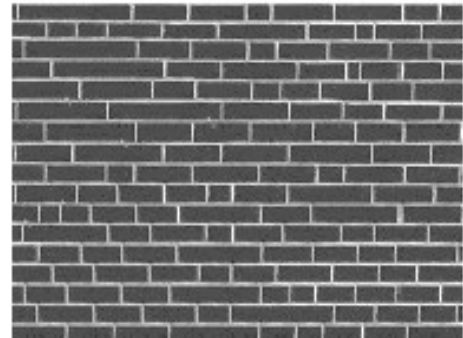
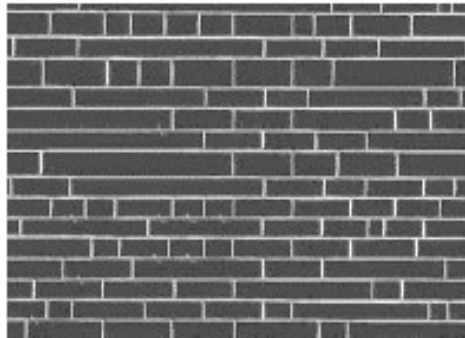
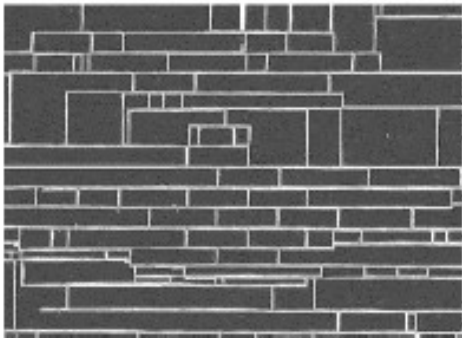
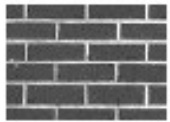
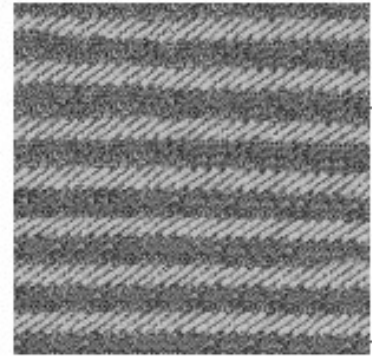
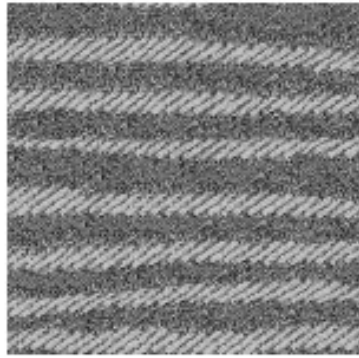
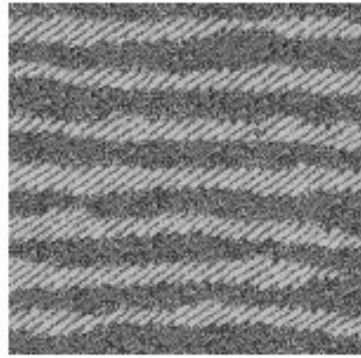
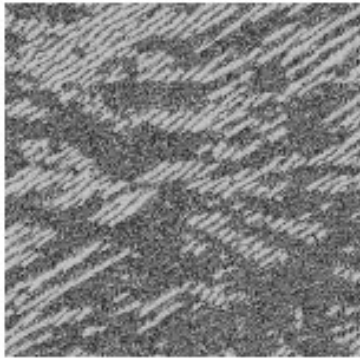
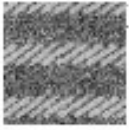
Erreur minimale

Influence de la taille de l'échantillon



<http://www.cs.berkeley.edu/~efros/research/NPS/efros-iccv99.ppt>

Résultats de synthèse de texture

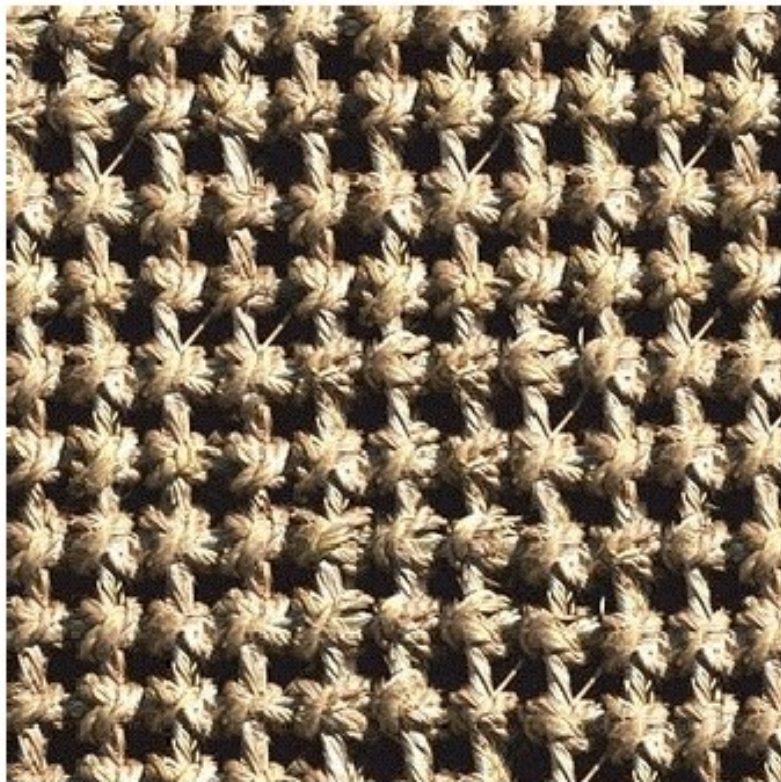


Taille de l'échantillon



<http://www.cs.berkeley.edu/~efros/research/NPS/efros-iccv99.ppt>

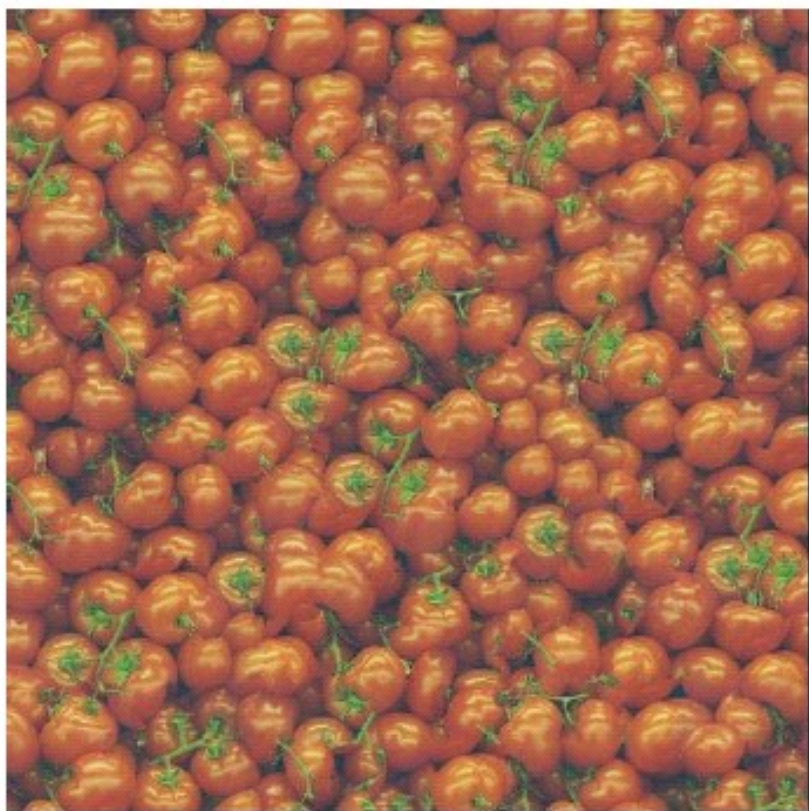
Source : David Lowe, *Texture*, CPSC 425: Computer Vision, UBC (Canada).



Source : David Lowe, *Texture*, CPSC 425: Computer Vision, UBC (Canada).



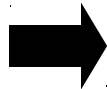
Problèmes



Source : David Lowe, *Texture*, CPSC 425: Computer Vision, UBC (Canada).

Synthèse de texte à partir d'images

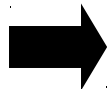
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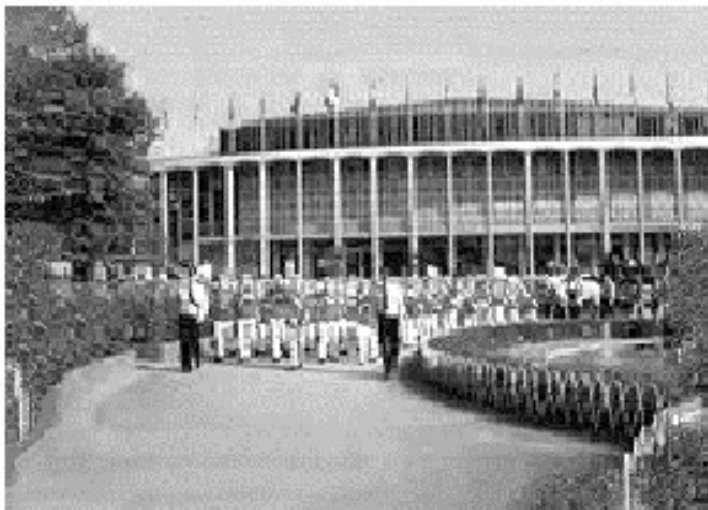
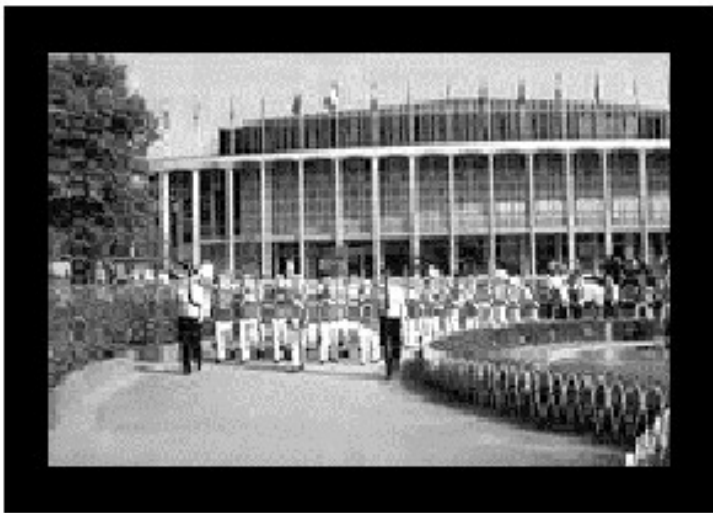
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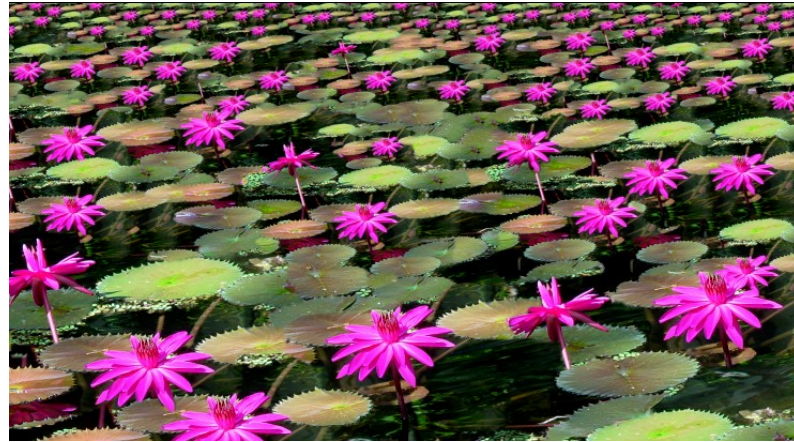
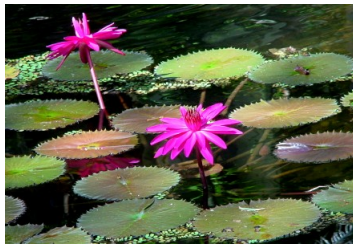
Extrapolation d'images



Extrapolation d'images



Extrapolation d'images





Références

(voir aussi la page web du cours)

- Stina Svensson, Representation and Description, Computerized image analysis, Swedish University of Agricultural Sciences.
 - www.cb.uu.se/~stina/bildSLU/lectures/F7_8vt2005.pdf
- Cornelia Fermüller, Lecture 24: Texture, CMSC 426 Image Processing (Computer Vision), University of Maryland at College Park.
 - <http://www.cfar.umd.edu/~fer/cmssc426/lectures/Texture2.ppt>
- Christopher Rasmussen, Texture, CIS 489/689 Computer Vision, University of Delaware (USA).
 - http://vision.cis.udel.edu/cv/lectures/cv19_texture.ppt



Références

(voir aussi la page web du cours)

- David Lowe, Texture, CPSC 425: Computer Vision, University of British Columbia (Canada).
 - <http://www.cs.ubc.ca/~lowe/425/slides/5-Texture.ppt>
- Frank Dellaert, Texture, CS x495 Computer Vision, Georgia Institute of Technology (USA).
 - http://www-static.cc.gatech.edu/classes/AY2005/cs4495_fall/09-Texture.ppt
- Marc Pollefeys, Class 7 Pyramids and textures, Comp256 Computer Vision, University of North Carolina at Chapel Hill (USA).
 - <http://www.cs.unc.edu/Research/vision/comp256/vision07.ppt>